

Smart Traffic Lights Using Local AI

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Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. Smart systems can use camera information and artificial intelligence to estimate traffic conditions and adjust signal timing. This paper discusses a local-AI-based traffic-light architecture in which traffic information is processed near the road to reduce communication delay and internet data usage. The proposed system also considers emergency vehicle detection as an important research direction.

Index Terms—Smart traffic lights, artificial intelligence, local AI, machine vision, traffic signal control, emergency vehicles.

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded.

Recent systems use cameras and artificial intelligence (AI) to estimate the number of vehicles present and then adjust the lights. Instead of changing lights only according to a fixed clock, the system uses information obtained from the road.

Research reviews show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

II. LOCAL AI TRAFFIC SYSTEM

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay.

Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions.

A small local computer can examine camera input and send a short instruction to the traffic-light controller.

Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

III. SYSTEM COMPARISON

The three systems considered in this paper are basic timed lights, cloud AI systems, and local smart cameras.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision Time	Internet Data	Setup Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI Systems	2 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

The comparison indicates that local smart cameras can provide faster decision making than basic timed lights while using less internet data than cloud-based systems.

IV. SYSTEM ARCHITECTURE

The proposed architecture follows four main steps:

- 1) Street cameras capture traffic conditions.
- 2) Smart pole computers perform local processing.
- 3) Selected information can be forwarded to a city cloud server.
- 4) The traffic-light controller changes the signals according to the decision.

The overall flow is:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights (1)

V. SYSTEM FLOW

The operation of the proposed system can be represented by the following flow:

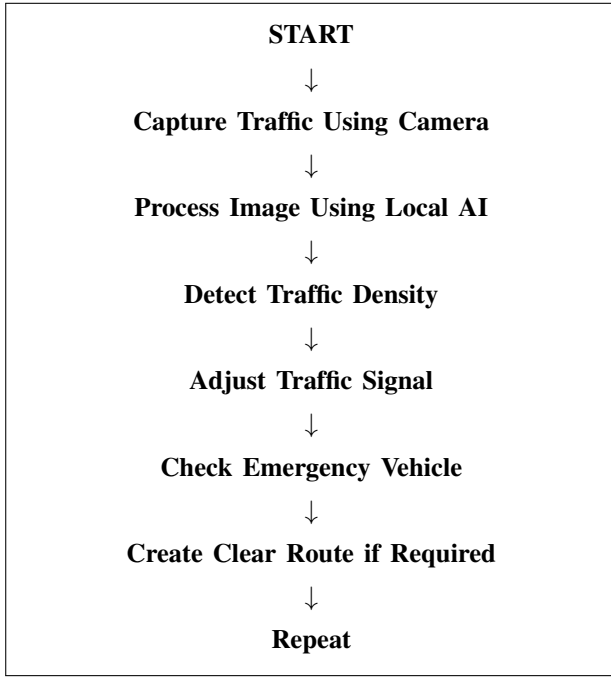


Fig. 1. Flowchart of the proposed local-AI traffic system

VI. RESEARCH GAP

Existing traffic systems can work well for regular daily traffic, especially when traffic patterns are predictable. However, an important emergency problem remains: many systems do not have a built-in way to automatically clear a path for a speeding ambulance.

This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route.

VII. EMERGENCY VEHICLE PRIORITY

An important extension of the proposed system is emergency vehicle priority.

The local AI system can analyze camera input to identify an ambulance or another emergency vehicle. When an emergency vehicle is detected, the system can communicate with nearby traffic signals and provide a clear route.

The basic decision process can be expressed as:

$$D = \begin{cases} \text{Normal Signal Control,} & E = 0 \\ \text{Emergency Priority,} & E = 1 \end{cases} \quad (2)$$

where E represents the detection status of an emergency vehicle.

VIII. ADVANTAGES

The proposed local-AI approach provides several possible advantages:

- Faster local decision making.
- Reduced dependence on continuous internet connectivity.
- Lower amount of traffic information transmitted to the cloud.
- Adaptive control based on actual road conditions.
- Possibility of emergency vehicle priority.

IX. APPLICATIONS

The proposed system can be applied in:

- Urban road intersections.
- High-traffic junctions.
- Emergency vehicle routes.
- Smart-city traffic management.
- Traffic monitoring systems.

X. CONCLUSION

This paper presents a simple local-AI-based architecture for smart traffic-light control. Instead of depending completely on fixed timers or cloud processing, local computers can process traffic information near the road and make faster decisions.

Research on AI-based traffic signal control and machine-vision-based adaptive traffic systems provides a basis for this approach [1], [2].

An important future direction is the detection of emergency vehicles and coordination of multiple traffic signals to provide a clear path for ambulances.

REFERENCES

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